

~~$P(\text{Alarm} = \text{True})$~~

Stage 2 - Bababeyk

~~$P(\text{Alarm} = \text{True})$~~ ~~$P(\text{Alarm} = \text{True}, \text{Burglary} = \text{false})$~~

①

$P(\text{Alarm} = \text{True}, \text{Burglary} = \text{false}, \text{Earthquake} = \text{false},$
 $\text{John Calls} = \text{True}, \text{Mary Calls} = \text{True})$



~~$P(A \sim B)$~~

$$P(A, \sim B, \sim E, J, M)$$

$$= P(\sim B) \cdot P(\sim E) \cdot P(A | \sim B, \sim E) P(J | A) \cdot P(M | A)$$

$$P(B) = 0.001 \Rightarrow P(\sim B) = 0.999$$

$$P(E) = 0.002 \Rightarrow P(\sim E) = 0.998$$

$$P(A | B=F, E=F) = 0.001$$

$$P(J | A=T) = 0.9$$

$$P(M | A=T) = 0.1$$

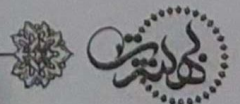
$$\Rightarrow P = 0.999 \times 0.998 \times 0.001 \times 0.9 \times 0.1$$

$$\approx 0.0009281 \approx 0.000928$$

① $P(\text{John calls})$

$$P(J) = P(J | A) P(A) + P(J | \sim A) P(\sim A)$$

$$P(A) = \sum_{b, e} P(b) P(e) P(A | b, e)$$



محاسبه احتمال برابر بوی

1. $B = T, E = T$

$$0.1001 \times 0.002 \times 0.99 = 0.00019982$$

2. $B = T, E = F$

$$0.1001 \times 0.999 \times 0.01 \approx 0.000999$$

3. $B = F, E = T$

$$0.999 \times 0.002 \times 0.29 \approx 0.000579$$

4. $B = F, E = F$

$$0.999 \times 0.999 \times 0.001 \approx 0.000998$$

$$\Rightarrow P(A) = 0.00019982 + 0.000999 + 0.000579 + 0.000998 \approx 0.002776$$

$$\Rightarrow P(\sim A) = 1 - 0.002776 = 0.997224$$

$$\Rightarrow P(J) = 0.9 \times 0.002776 + 0.1 \times 0.997224$$

$$= 0.002501 + 0.0997224 \approx 0.102223$$



٥) $P(\text{Burglary} | \text{John calls, Mary calls})$

$$P(B | J, M) = \frac{P(B, J, M)}{P(J, M)}$$

$$P(B=T, J, M) = \sum_E \sum_A P(B) P(E) P(A|B, E) P(J|A) P(M|A)$$

case 1: $B=T, E=T$

$$P = 0,001 \times 0,002 \times 0,99 \times 0,9 \times 0,1 \checkmark$$

$$\approx 0,0000119 \checkmark$$

case 2: $B=T, E=F$

$$P = 0,001 \times 0,999 \times 0,99 \times 0,9 \times 0,1 \checkmark$$

$$= 0,000099$$

$$\Rightarrow P(B, J, M) = 0,0000119 \checkmark + 0,000099$$

$$\approx 0,0000991 \checkmark$$

$$P(J, M) = \sum_b \sum_e \sum_a P(b) P(e) P(a|b, e) P(J|a) P(M|a)$$

$$P(B=T) = 0,001, P(B=F) = 0,999$$

$$P(E=T) = 0,002, P(E=F) = 0,998$$



B	E	$P(A=T B, E)$	$P(A=F B, E)$
T	T	0,90	0,00
T	F	0,94	0,04
F	T	0,29	0,71
F	F	0,001	0,999

$$P(J=T|A=T) = 0,9$$

$$P(J=T|A=F) = 0,00$$

$$P(M=T|A=T) = 0,7$$

$$P(M=T|A=F) = 0,01$$

$$P(J, M|b, e) = P(A=T|b, e) \cdot P(J=T|A=T)$$

$$+ P(A=F|b, e) \cdot P(J=T|A=F)$$

$$= P(M=T|A=T)$$

$$P(J, M|A=T) = 0,9 \times 0,7 = 0,63$$

$$P(J, M|A=F) = 0,00 \times 0,01 = 0,0000$$

(B, E)

⊕ B = T, E = T

$$\begin{aligned} P(J, M | B, E) &= 0,95 \times 0,55 + 0,05 \times 0,0001 \\ &= \cancel{0,95} \times 0,55 + 0,00005 \\ &= 0,522505 \end{aligned}$$

$$\begin{aligned} P(B, E, J, M) &= 0,001 \times 0,0005 \times 0,522505 \\ &= 0,0000002612525 \end{aligned}$$

B = T, E = F

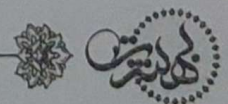
$$\begin{aligned} P(J, M | B, E) &= 0,95 \times 0,55 + 0,05 \times 0,0001 \\ &= 0,522505 + 0,00005 \\ &= 0,522555 \end{aligned}$$

$$\begin{aligned} P(B, E, J, M) &= 0,001 \times 0,998 \times 0,522555 \\ &\approx 0,0005214199 \end{aligned}$$

B = F, E = T

$$\begin{aligned} P(J, M | B, E) &= 0,29 \times 0,55 + 0,71 \times 0,0001 \\ &= 0,1595 + 0,000071 \\ &= 0,159571 \end{aligned}$$

$$\begin{aligned} P(B, E, J, M) &= 0,999 \times 0,0005 \times 0,159571 \\ &\approx 0,0000795995 \end{aligned}$$



$$B = F, E = F$$

$$P(J, M | B, E) = 0,001 \times 0,5^5 + 0,999 \times 0,000$$

$$= 0,0003125 + 0,0000000$$

$$= 0,0003125$$

$$P(B, E, J, M) = 0,999 \times 0,999 \times 0,0003125$$

$$\approx 0,0003125$$

$$P(J, M) = 0,0000001195 + 0,000000001 + 0,000000005 + 0,0003125 = 0,0003126$$

$$\Rightarrow P(B | J, M) = \frac{P(B, J, M)}{P(J, M)}$$

$$= \frac{0,0000001195}{0,0003126} \approx 0,000382$$